THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET B

Chapter 13: **Two Variables, One Line** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of solutions of a linear equation in two variables is: 1

(a) exactly one **(b)** zero **(c)** infinite **(d)** exactly two

2. A pair of linear equations has no solution when their graphs are: 1

(a) parallel lines **(b)** perpendicular lines only

(c) intersecting lines **(d)** coincident lines

3. **Assertion (A):** The pair 2x + 3y = 5 and 4x + 6y = 11 has no solution. 1

Reason (R): If a1/a2 = b1/b2 ≠ c1/c2, the pair of equations has no solution.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

4. The solution of the pair x + y = 5 and x − y = 1 is: 1

(a) (-3, 2) **(b)** (2, 3) **(c)** (3, -2) **(d)** (3, 2)

5. If x and y satisfy x + 2y = 8 and 3x − y = 17, then the value of x − y is: 1

(a) 3 **(b)** 8 **(c)** 6 **(d)** 5

6. The slope of the line through the points (0, -3) and (4, 5) is: 1

(a) 1/2 **(b)** 3 **(c)** -2 **(d)** 2

7. The y-intercept of the line 5x + 2y = 8 is: 1

(a) -8/5 **(b)** -4 **(c)** 8/5 **(d)** 4

8. Which of the following points lies on the line 4x + 3y = 10? 1

(a) (1, 3) **(b)** (1, 2) **(c)** (2, 2) **(d)** (2, 3)

9. The pair of equations 5x − 10y = 15 and x − 2y = 3 has: 1

(a) exactly two solutions **(b)** no solution

(c) a unique solution **(d)** infinitely many solutions

10. Two numbers have sum 30 and difference 8. The larger number is: 1

(a) 11 **(b)** 19 **(c)** 38 **(d)** 22

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the slope of the line through $(-2, 5)$ and $(4, -1)$. 2

12. Write $4x + 2y = 10$ in slope-intercept form and read off the slope and the y-intercept. 2

13. State the slope of the lines 2
(i) $y = -3$ and
(ii) $x = 2$.

14. Why does the line $5x - 2y = 0$ pass through the origin? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Without solving, state whether each pair has a unique solution, no solution or infinitely many solutions: 3
(i) $2x - 3y + 5 = 0,\ 4x - 6y + 10 = 0$;
(ii) $3x + 2y - 7 = 0,\ 6x + 4y - 9 = 0$;
(iii) $x + 2y - 5 = 0,\ 2x - y + 3 = 0$.

16. Find the value of k for which the pair $kx + 2y = 4$ and $3x + 6y = 12$ has infinitely many solutions. Why does this value not give a unique solution? 3

17. A line passes through $(1, 3)$ and $(3, 7)$. Find its slope and its equation in the form $y = mx + d$. 3

OR

Solve graphically: $x + y = 6$ and $x - y = 2$. Write a table of two points for each line and state the nature of the solution. 3

18. Half the perimeter of a rectangular garden is 30 m, and its length is 6 m more than its width. Find its length and width. 3

OR

Plan A costs ₹60 per month plus ₹2 per minute; Plan B costs ₹20 per month plus ₹4 per minute. For how many minutes do the two plans cost the same, and for what usage is Plan A cheaper? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The ratio of the incomes of two persons is 5 : 4 and the ratio of their expenditures is 3 : 2. Each saves ₹3000 per month. Find their monthly incomes and expenditures. 5

OR

A cricket coach buys 5 bats and 8 balls for ₹9300 and later buys 3 bats and 6 balls for ₹6000. Form a pair of equations and find the cost of one bat and one ball. 5

20. A library charges a fixed amount for the first 3 days and an extra amount for each day after that. Rohan paid ₹36 for 7 days and Sana paid ₹24 for 4 days. Find the fixed charge and the charge per extra day, and the cost of keeping a book for 10 days. 5

OR

Case study: A school fair has two stalls. Stall P charges ₹20 entry plus ₹5 per game; Stall Q charges ₹5 entry plus ₹10 per game. 5
(i) Write the cost of g games at each stall.
(ii) For how many games is the total cost the same?
(iii) Which stall is cheaper for 2 games?
(iv) Represent this using the slopes and intercepts of the two lines.

END OF QUESTION PAPER — **Total: 40 Marks**